Did the native reranker improve over official frozen R77B?

Split	n	R77B Top1 %	R124 Top1 %	Delta pp	R77B regret %	R124 regret %
TEST	81000	71.2593	71.3284	0.0691	0.3944	0.3962
VAL	81000	71.0358	71.0185	-0.0173	0.4030	0.4071

Single VAL lock update 5760; checkpoint 5477aebc3d252583.
ALL is equal-task. Both NEW4500 populations and FP64 cost sources match exactly; old3895 metrics are never reused.
Success is native correction minus damage and actual regret, NOT auxiliary quality loss. TEST was exposed during research.
R77B is a complete external pretrained selector; the new original4layer128 encoder is scratch-initialized.
MetaTRAIN10886/task is source-record unseen only; no exhaustive content/base-copy certificate.

NSS / official R77B / SAME locked R124: four matched datasets

Dataset	n	NSS Top1 %	R77B Top1 %	R124 Top1 %	NSS mean cost	R77B cost	R124 cost	NSS status
TSP TEST	4500	45.8889	46.6444	46.6444	7.9704	7.9711	7.9716	complete
CVRP TEST	4500	45.9778	47.2222	47.8000	18.0048	18.0122	18.0142	complete
TSPLIB	49	20.4082	20.4082	16.3265	8.2179	8.2167	8.2203	complete
CVRPLIB	100	70.0000	60.0000	62.0000	67.1011	67.3003	67.2938	complete

NSS: historical R79 checkpoints, no new90k retraining. TEST4500 is the current release; LIB is original native LIB.
All columns use saved native winner and full FP64 cost alignment. Missing references remain MISSING, never synthesized.
Release: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018
TSP pool: BQ, DIFUSCO, DIFUSCO500, ELG, LEHD, OMNI, T2T, T2T500
CVRP pool: BQ, ELG, ICAM, LEHD, MVMOE, MoSES_CaDA, MoSES_RF, OMNI, RELD_CVRP, RouteFinder

VAL: ALL and all18 tasks

Task	n	Top1 %	mean cost	SBS cost	Oracle	vs SBS %	vs Oracle %	actual regret %
ALL	81000	71.0185	11.4170	11.4280	11.3717	-0.1918	0.3959	0.4071
TSP	4500	46.6222	8.0254	8.0454	7.9905	-0.2478	0.4366	0.4289
CVRP	4500	46.2889	18.1630	18.2694	18.0641	-0.5822	0.5476	0.7327
ATSP	4500	77.3556	1.6385	1.6769	1.6325	-2.2877	0.3719	0.3695
OVRP	4500	41.5333	7.4613	7.4696	7.4158	-0.1104	0.6143	0.6275
VRPB	4500	81.0667	9.3541	9.3549	9.3187	-0.0085	0.3791	0.3882
VRPL	4500	37.7111	12.4311	12.4380	12.3834	-0.0550	0.3850	0.3928
VRPTW	4500	80.0000	18.8190	18.8194	18.7661	-0.0021	0.2819	0.2982
OVRPTW	4500	83.8444	10.1431	10.1437	10.1169	-0.0061	0.2595	0.2562
OVRPB	4500	88.5111	6.3573	6.3573	6.3411	0.0000	0.2551	0.2650
OVRPL	4500	40.1778	7.4393	7.4457	7.3904	-0.0855	0.6620	0.6860
VRPBL	4500	78.7556	9.3690	9.3715	9.3284	-0.0273	0.4352	0.4376
VRPBTW	4500	78.5778	19.9472	19.9504	19.8511	-0.0160	0.4843	0.4637
VRPLTW	4500	78.8667	18.7646	18.7659	18.7078	-0.0066	0.3036	0.3188
OVRPBL	4500	89.2444	6.3698	6.3701	6.3528	-0.0051	0.2667	0.2780
OVRPBTW	4500	83.6000	10.6603	10.6611	10.6232	-0.0070	0.3493	0.3221
OVRPLTW	4500	83.4889	10.1575	10.1574	10.1298	0.0010	0.2735	0.2782
VRPBLTW	4500	79.4444	19.7957	19.7970	19.7064	-0.0069	0.4530	0.4404
OVRPBLTW	4500	83.2444	10.6101	10.6100	10.5713	0.0014	0.3671	0.3449

SBS fitted on frozen TRAIN81000. Oracle = each instance legal-pool minimum.
vs_SBS/vs_Oracle are ratios of mean costs; actual regret is the mean per-instance relative regret.
ALL averages tasks equally; raw-cost macro means are descriptive, not cross-task optimization evidence.

VAL: official R77B reference on the SAME new4500

Task	R77B Top1 %	R124 Top1 %	Delta pp	R77B mean cost	R124 mean cost	R77B regret %	R124 regret %
ALL	71.0358	71.0185	-0.0173	11.4167	11.4170	0.4030	0.4071
TSP	46.7778	46.6222	-0.1556	8.0254	8.0254	0.4251	0.4289
CVRP	46.5556	46.2889	-0.2667	18.1583	18.1630	0.6879	0.7327
ATSP	77.4222	77.3556	-0.0667	1.6385	1.6385	0.3672	0.3695
OVRP	40.9556	41.5333	0.5778	7.4619	7.4613	0.6347	0.6275
VRPB	81.0667	81.0667	0.0000	9.3537	9.3541	0.3809	0.3882
VRPL	38.1111	37.7111	-0.4000	12.4313	12.4311	0.3954	0.3928
VRPTW	80.0444	80.0000	-0.0444	18.8188	18.8190	0.2967	0.2982
OVRPTW	83.8667	83.8444	-0.0222	10.1431	10.1431	0.2558	0.2562
OVRPB	88.5111	88.5111	0.0000	6.3573	6.3573	0.2650	0.2650
OVRPL	40.1778	40.1778	0.0000	7.4384	7.4393	0.6658	0.6860
VRPBL	78.8889	78.7556	-0.1333	9.3683	9.3690	0.4305	0.4376
VRPBTW	78.5333	78.5778	0.0444	19.9472	19.9472	0.4626	0.4637
VRPLTW	78.8444	78.8667	0.0222	18.7647	18.7646	0.3196	0.3188
OVRPBL	89.2444	89.2444	0.0000	6.3698	6.3698	0.2780	0.2780
OVRPBTW	83.5556	83.6000	0.0444	10.6604	10.6603	0.3228	0.3221
OVRPLTW	83.4889	83.4889	0.0000	10.1575	10.1575	0.2782	0.2782
VRPBLTW	79.3556	79.4444	0.0889	19.7960	19.7957	0.4416	0.4404
OVRPBLTW	83.2444	83.2444	0.0000	10.6102	10.6101	0.3463	0.3449

VAL f0 scores verified against independent shared R123 R77B archives; TEST f0 scores from the same post-lock full forward.
Official frozen R77B checkpoint update34776, SHA 31c0c75d66b1bde6ad533909a9c04b0036e80af4c4a5775b01b957ce310e6d1b. No old3895 accuracy/cost summaries.

TEST: ALL and all18 tasks

Task	n	Top1 %	mean cost	SBS cost	Oracle	vs SBS %	vs Oracle %	actual regret %
ALL	81000	71.3284	11.3741	11.3853	11.3299	-0.1961	0.3871	0.3962
TSP	4500	46.6444	7.9716	7.9896	7.9376	-0.2250	0.4290	0.4228
CVRP	4500	47.8000	18.0142	18.1250	17.9185	-0.6114	0.5338	0.7034
ATSP	4500	76.8444	1.6353	1.6742	1.6294	-2.3268	0.3566	0.3573
OVRP	4500	40.7556	7.4467	7.4557	7.4005	-0.1212	0.6247	0.6355
VRPB	4500	80.5778	9.3504	9.3518	9.3153	-0.0152	0.3767	0.3790
VRPL	4500	37.4000	12.3488	12.3561	12.3010	-0.0592	0.3891	0.3962
VRPTW	4500	81.0000	18.6897	18.6905	18.6374	-0.0043	0.2808	0.2917
OVRPTW	4500	83.9333	10.0927	10.0931	10.0681	-0.0032	0.2444	0.2357
OVRPB	4500	88.0889	6.3711	6.3708	6.3522	0.0043	0.2971	0.3049
OVRPL	4500	42.7333	7.4631	7.4725	7.4180	-0.1269	0.6070	0.6183
VRPBL	4500	79.6222	9.3949	9.3964	9.3560	-0.0155	0.4164	0.4200
VRPBTW	4500	78.8667	19.6639	19.6657	19.5777	-0.0092	0.4406	0.4369
VRPLTW	4500	79.2444	18.8024	18.8031	18.7451	-0.0040	0.3054	0.3225
OVRPBL	4500	89.2667	6.3758	6.3757	6.3593	0.0012	0.2596	0.2710
OVRPBTW	4500	84.8222	10.6596	10.6596	10.6247	-0.0003	0.3276	0.3003
OVRPLTW	4500	82.8889	10.0858	10.0861	10.0590	-0.0032	0.2663	0.2678
VRPBLTW	4500	79.8222	19.7054	19.7071	19.6133	-0.0086	0.4695	0.4485
OVRPBLTW	4500	83.6000	10.6621	10.6621	10.6256	-0.0007	0.3436	0.3208

SBS fitted on frozen TRAIN81000. Oracle = each instance legal-pool minimum.
vs_SBS/vs_Oracle are ratios of mean costs; actual regret is the mean per-instance relative regret.
ALL averages tasks equally; raw-cost macro means are descriptive, not cross-task optimization evidence.

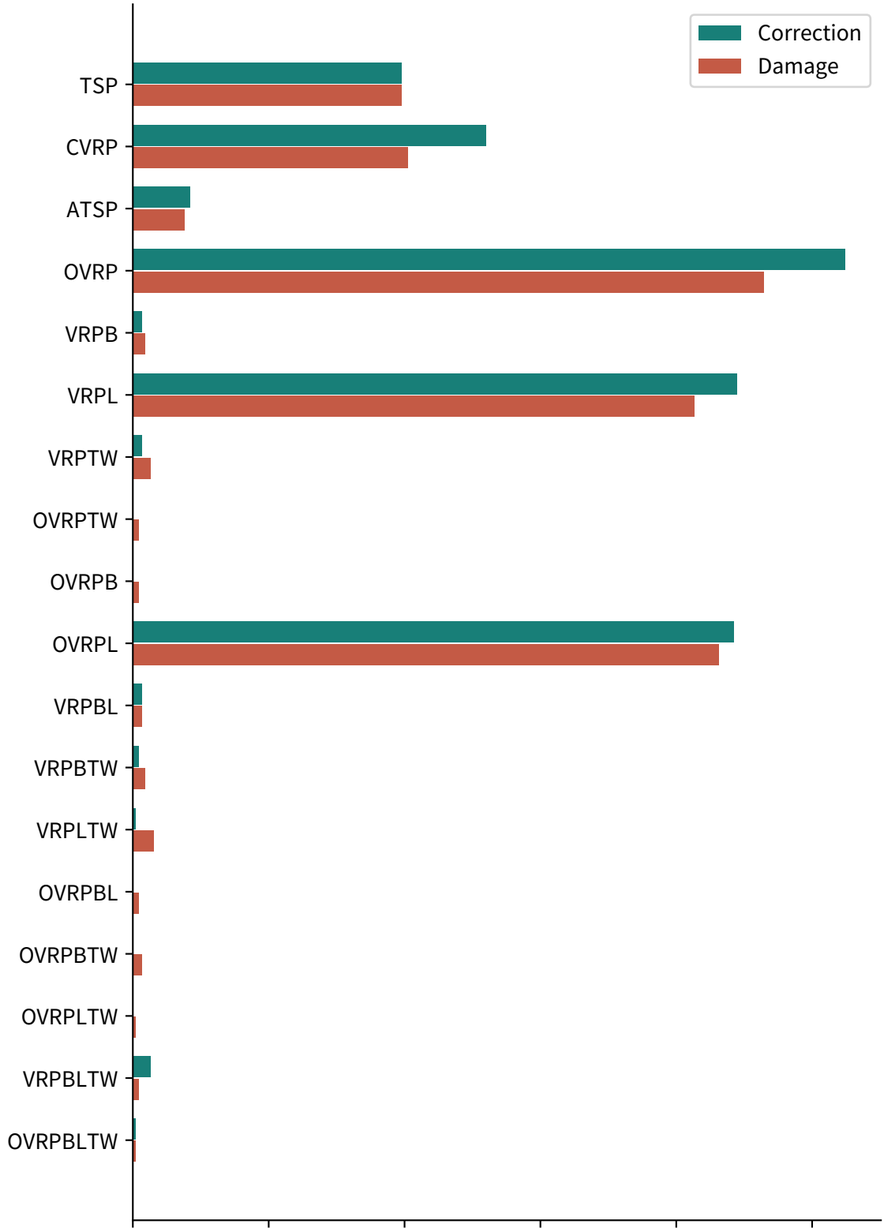
TEST: official R77B reference on the SAME new4500

Task	R77B Top1 %	R124 Top1 %	Delta pp	R77B mean cost	R124 mean cost	R77B regret %	R124 regret %
ALL	71.2593	71.3284	0.0691	11.3739	11.3741	0.3944	0.3962
TSP	46.6444	46.6444	0.0000	7.9711	7.9716	0.4151	0.4228
CVRP	47.2222	47.8000	0.5778	18.0122	18.0142	0.6843	0.7034
ATSP	76.8000	76.8444	0.0444	1.6354	1.6353	0.3682	0.3573
OVRP	40.1556	40.7556	0.6000	7.4473	7.4467	0.6410	0.6355
VRPB	80.6000	80.5778	-0.0222	9.3504	9.3504	0.3809	0.3790
VRPL	37.0889	37.4000	0.3111	12.3502	12.3488	0.4078	0.3962
VRPTW	81.0667	81.0000	-0.0667	18.6891	18.6897	0.2894	0.2917
OVRPTW	83.9778	83.9333	-0.0444	10.0927	10.0927	0.2348	0.2357
OVRPB	88.1333	88.0889	-0.0444	6.3707	6.3711	0.2980	0.3049
OVRPL	42.6222	42.7333	0.1111	7.4630	7.4631	0.6163	0.6183
VRPBL	79.6222	79.6222	0.0000	9.3944	9.3949	0.4100	0.4200
VRPBTW	78.9111	78.8667	-0.0444	19.6635	19.6639	0.4338	0.4369
VRPLTW	79.3778	79.2444	-0.1333	18.8015	18.8024	0.3175	0.3225
OVRPBL	89.3111	89.2667	-0.0444	6.3756	6.3758	0.2663	0.2710
OVRPBTW	84.8889	84.8222	-0.0667	10.6591	10.6596	0.2960	0.3003
OVRPLTW	82.9111	82.8889	-0.0222	10.0858	10.0858	0.2674	0.2678
VRPBLTW	79.7333	79.8222	0.0889	19.7059	19.7054	0.4513	0.4485
OVRPBLTW	83.6000	83.6000	0.0000	10.6622	10.6621	0.3214	0.3208

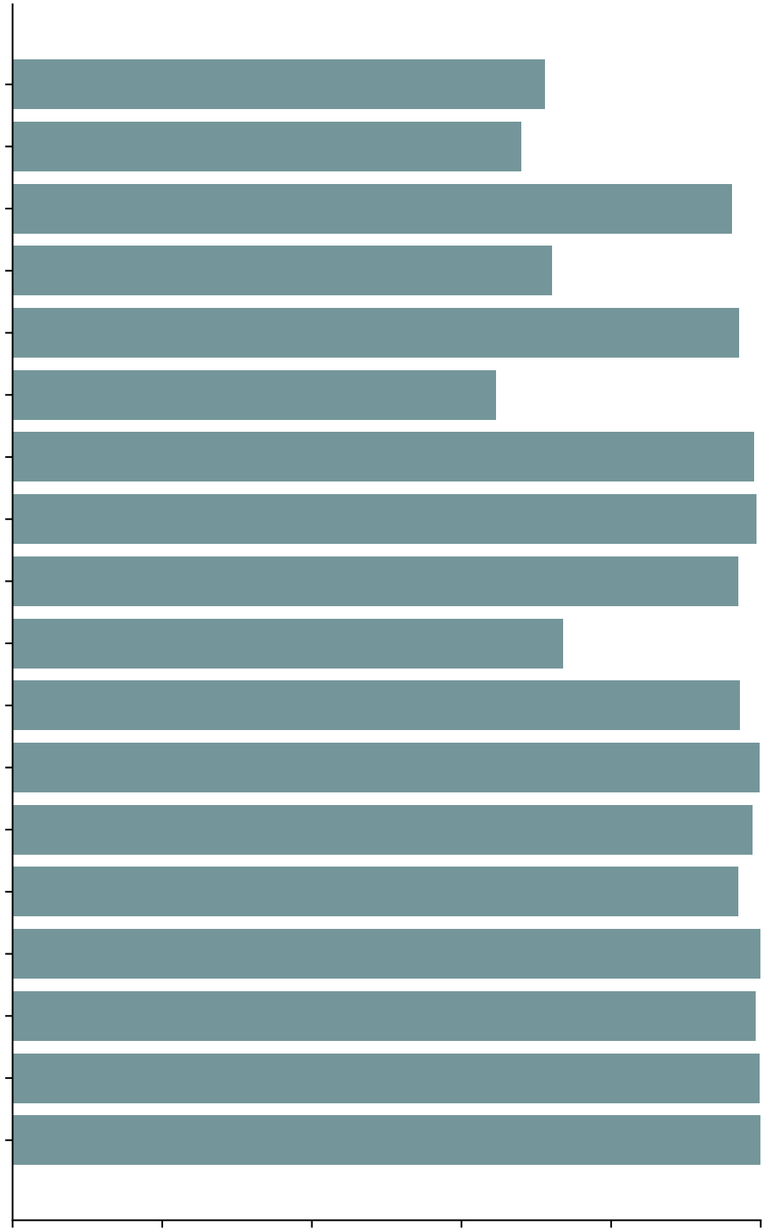
VAL f0 scores verified against independent shared R123 R77B archives; TEST f0 scores from the same post-lock full forward.
Official frozen R77B checkpoint update34776, SHA 31c0c75d66b1bde6ad533909a9c04b0036e80af4c4a5775b01b957ce310e6d1b. No old3895 accuracy/cost summaries.

Full18 TEST: what changed relative to official frozen R77B?

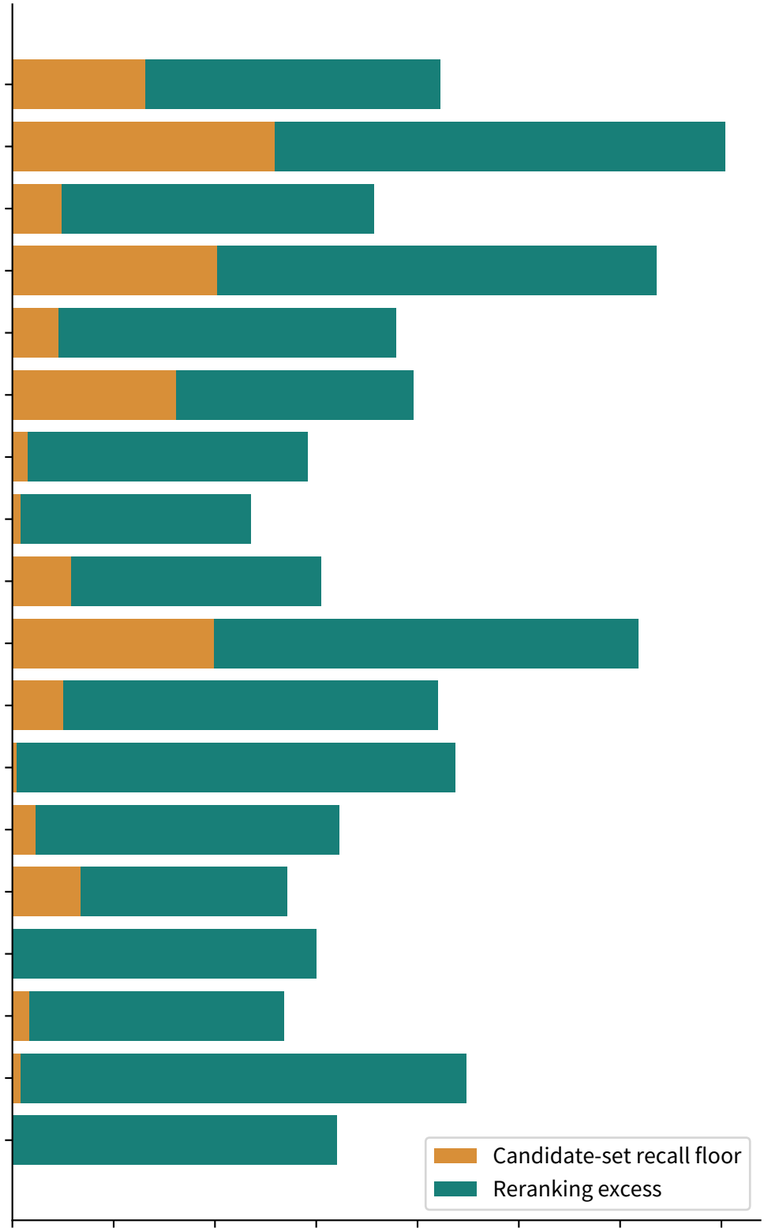
Net Top1 change = correction - damage



Coverage is a ceiling, not achieved accuracy



Two exact terms, SAME full legal Oracle



Full18 TEST: exact coverage and two-term regret accounting

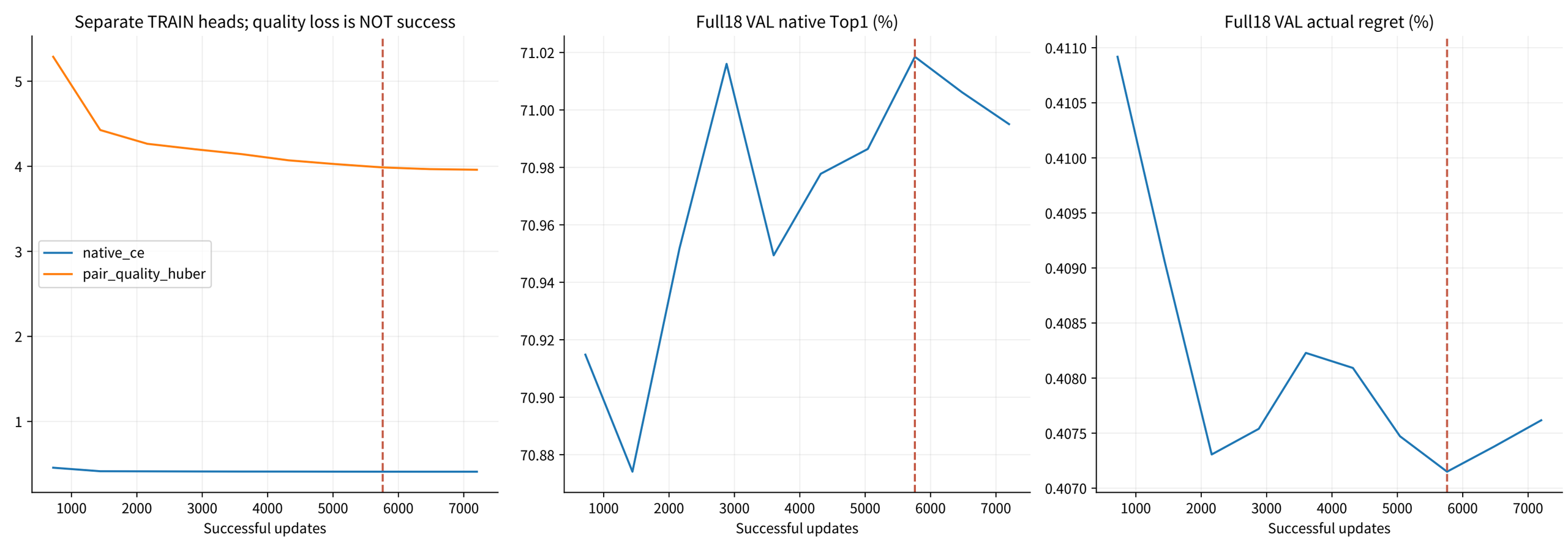
Task	Top2 cover %	Pair Top1 %	Corrected	Damaged	Net pp	Recall floor %	Excess %	Sum regret %
ALL	90.5963	78.7321	879	823	0.0691	0.0723	0.3239	0.3962
TSP	71.2000	65.5119	89	89	0.0000	0.1316	0.2912	0.4228
CVRP	68.0222	70.2712	117	91	0.5778	0.2593	0.4441	0.7034
ATSP	96.2000	79.8799	19	17	0.0444	0.0488	0.3085	0.3573
OVRP	72.0889	56.5351	236	209	0.6000	0.2020	0.4335	0.6355
VRPB	97.1333	82.9558	3	4	-0.0222	0.0462	0.3327	0.3790
VRPL	64.6222	57.8748	200	186	0.3111	0.1621	0.2341	0.3962
VRPTW	99.1333	81.7081	3	6	-0.0667	0.0157	0.2760	0.2917
OVRPTW	99.4444	84.4022	0	2	-0.0444	0.0086	0.2271	0.2357
OVRPB	97.0444	90.7717	0	2	-0.0444	0.0580	0.2469	0.3049
OVRPL	73.6222	58.0441	199	194	0.1111	0.1994	0.4189	0.6183
VRPBL	97.2444	81.8784	3	3	0.0000	0.0506	0.3694	0.4200
VRPBTW	99.8889	78.9544	2	4	-0.0444	0.0041	0.4328	0.4369
VRPLTW	98.9333	80.0988	1	7	-0.1333	0.0230	0.2995	0.3225
OVRPBL	97.0000	92.0275	0	2	-0.0444	0.0673	0.2037	0.2710
OVRPBTW	100.0000	84.8222	0	3	-0.0667	0.0000	0.3003	0.3003
OVRPLTW	99.3556	83.4265	0	1	-0.0222	0.0166	0.2513	0.2678
VRPBLTW	99.8000	79.9822	6	2	0.0889	0.0084	0.4401	0.4485
OVRPBLTW	100.0000	83.6000	1	1	0.0000	0.0000	0.3208	0.3208

Top1 = coverage * conditional accuracy. ALL conditional is macro joint-correct / macro coverage, NOT mean task conditional.
Regret = mean[(min pair cost - Oracle)/Oracle] + mean[(selected cost - min pair cost)/Oracle].
Native ties may cause a native error with zero cost regret. Pair-outside native winner is never a fake classification target.

Actual complete stage0 + reranker inference cost

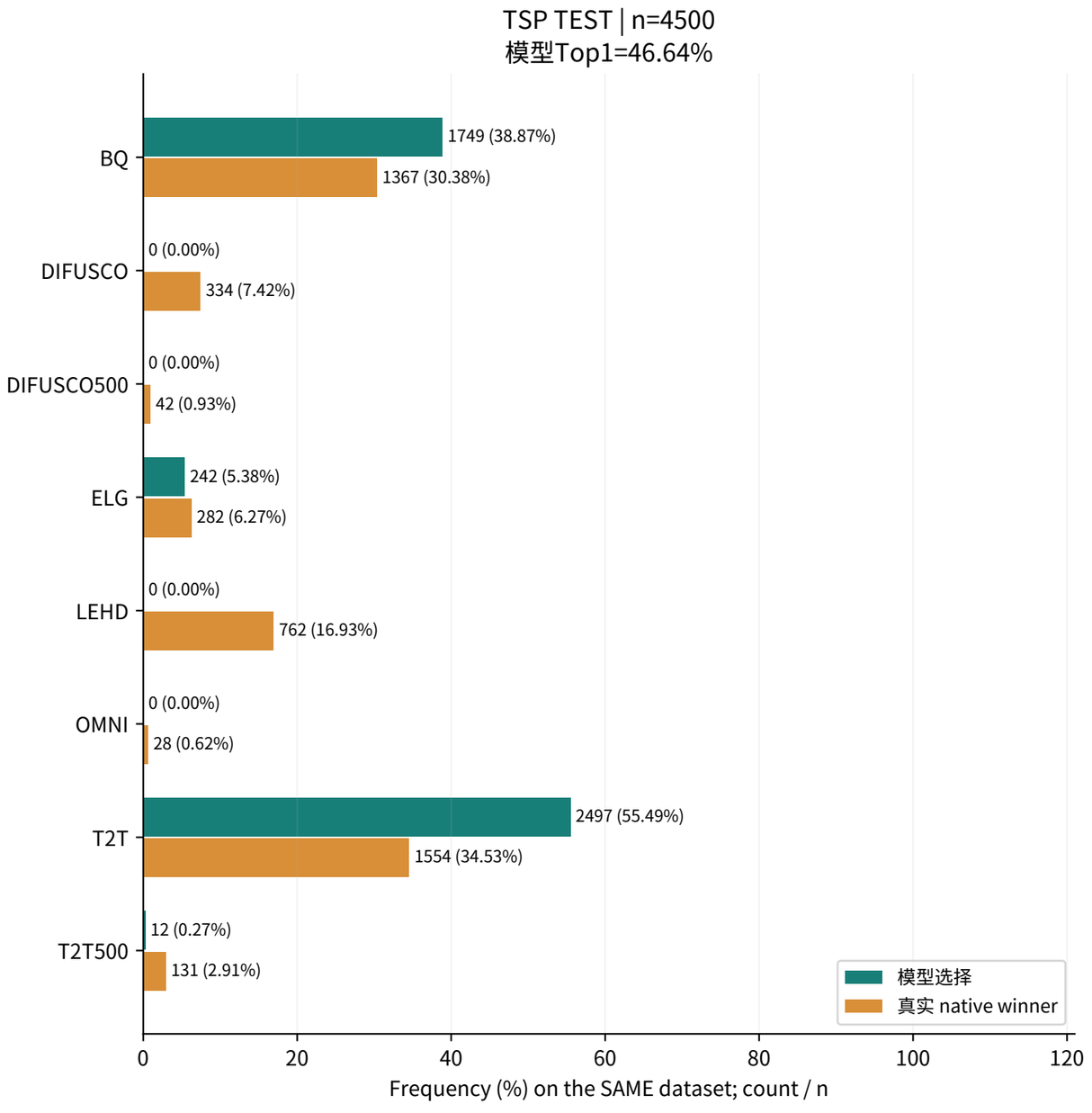
Dataset	Task	n	Full model sec	ms / instance	Cap	Full chain	Device	Geometry sec
TEST	ALL	81000	16.2573	0.2007	per-task	True	per-task	MISSING
TEST	TSP	4500	3.9704	0.8823	128	True	cuda:0	MISSING
TEST	CVRP	4500	4.2891	0.9531	128	True	cuda:0	MISSING
TEST	ATSP	4500	0.5121	0.1138	512	True	cuda:0	MISSING
TEST	OVRP	4500	0.4976	0.1106	512	True	cuda:0	MISSING
TEST	VRPB	4500	0.4978	0.1106	512	True	cuda:0	MISSING
TEST	VRPL	4500	0.4991	0.1109	512	True	cuda:0	MISSING
TEST	VRPTW	4500	0.4992	0.1109	512	True	cuda:0	MISSING
TEST	OVRPTW	4500	0.4998	0.1111	512	True	cuda:0	MISSING
TEST	OVRPB	4500	0.4990	0.1109	512	True	cuda:0	MISSING
TEST	OVRPL	4500	0.5009	0.1113	512	True	cuda:0	MISSING
TEST	VRPBL	4500	0.4989	0.1109	512	True	cuda:0	MISSING
TEST	VRPBTW	4500	0.4996	0.1110	512	True	cuda:0	MISSING
TEST	VRPLTW	4500	0.5012	0.1114	512	True	cuda:0	MISSING
TEST	OVRPBL	4500	0.4980	0.1107	512	True	cuda:0	MISSING
TEST	OVRPBTW	4500	0.4982	0.1107	512	True	cuda:0	MISSING
TEST	OVRPLTW	4500	0.5002	0.1111	512	True	cuda:0	MISSING
TEST	VRPBLTW	4500	0.4983	0.1107	512	True	cuda:0	MISSING
TEST	OVRPBLTW	4500	0.4980	0.1107	512	True	cuda:0	MISSING
TSPLIB	TSP	49	0.8708	17.7723	1	True	cuda:0	0.4094
CVRPLIB	CVRP	100	1.2773	12.7733	1	True	cuda:0	0.3166

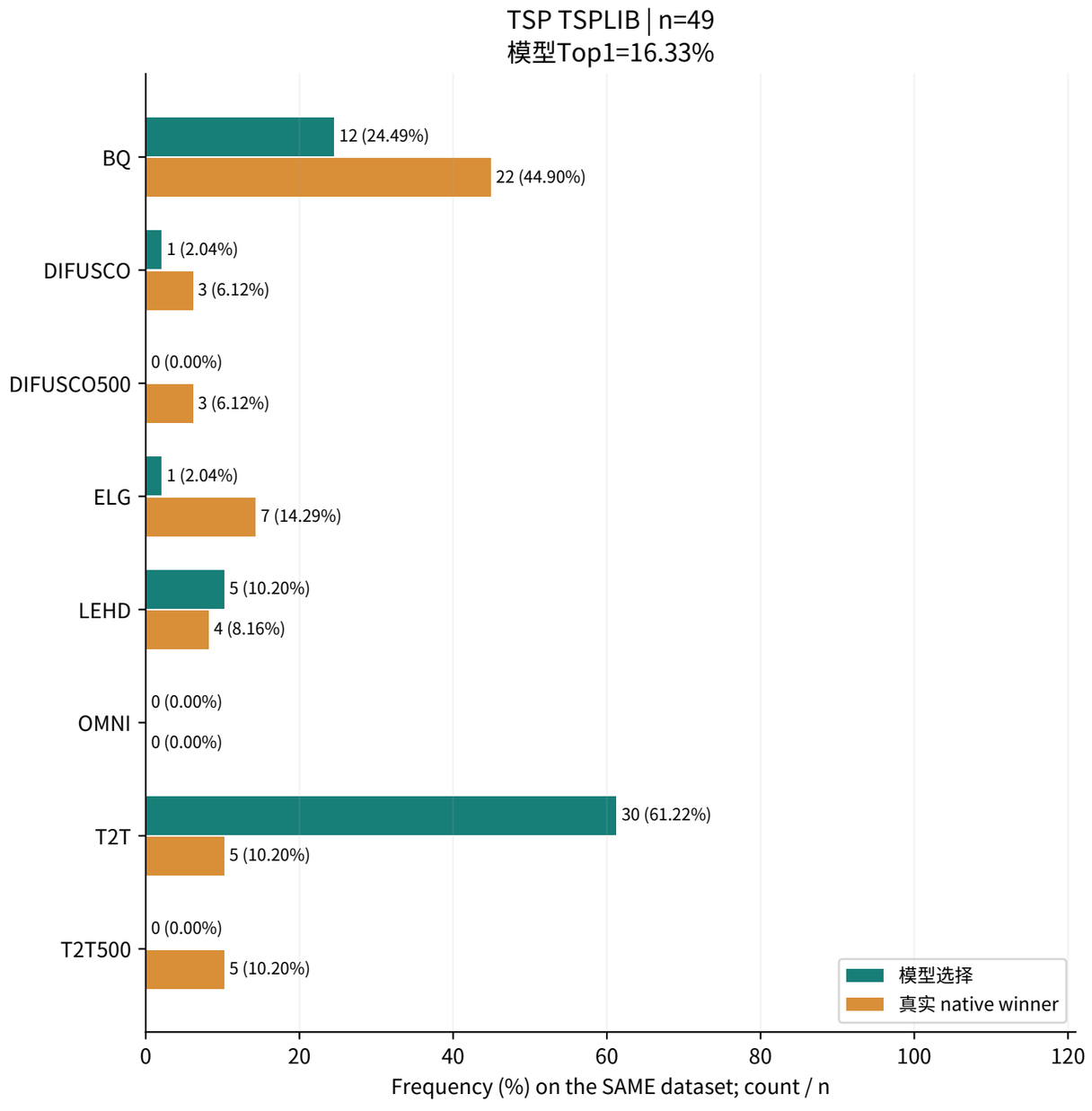
Every TEST/LIB model timing includes COMPLETE frozen R77B + new encoder + pair-native/quality heads; no readout-only estimate.
Actual first-pass FP32 batched forward, CUDA synchronized when applicable; no I/O, transfer or solver time.
VAL cached-f0 timing is NOT deployment cost. Shared bank precomputation is not free; LIB geometry is reported separately.



TRAIN loss = ALL-row mean [1(Y in Top2) native BCE] + Huber(quality/.01, pair-relative-quality/.01), coefficient 1.

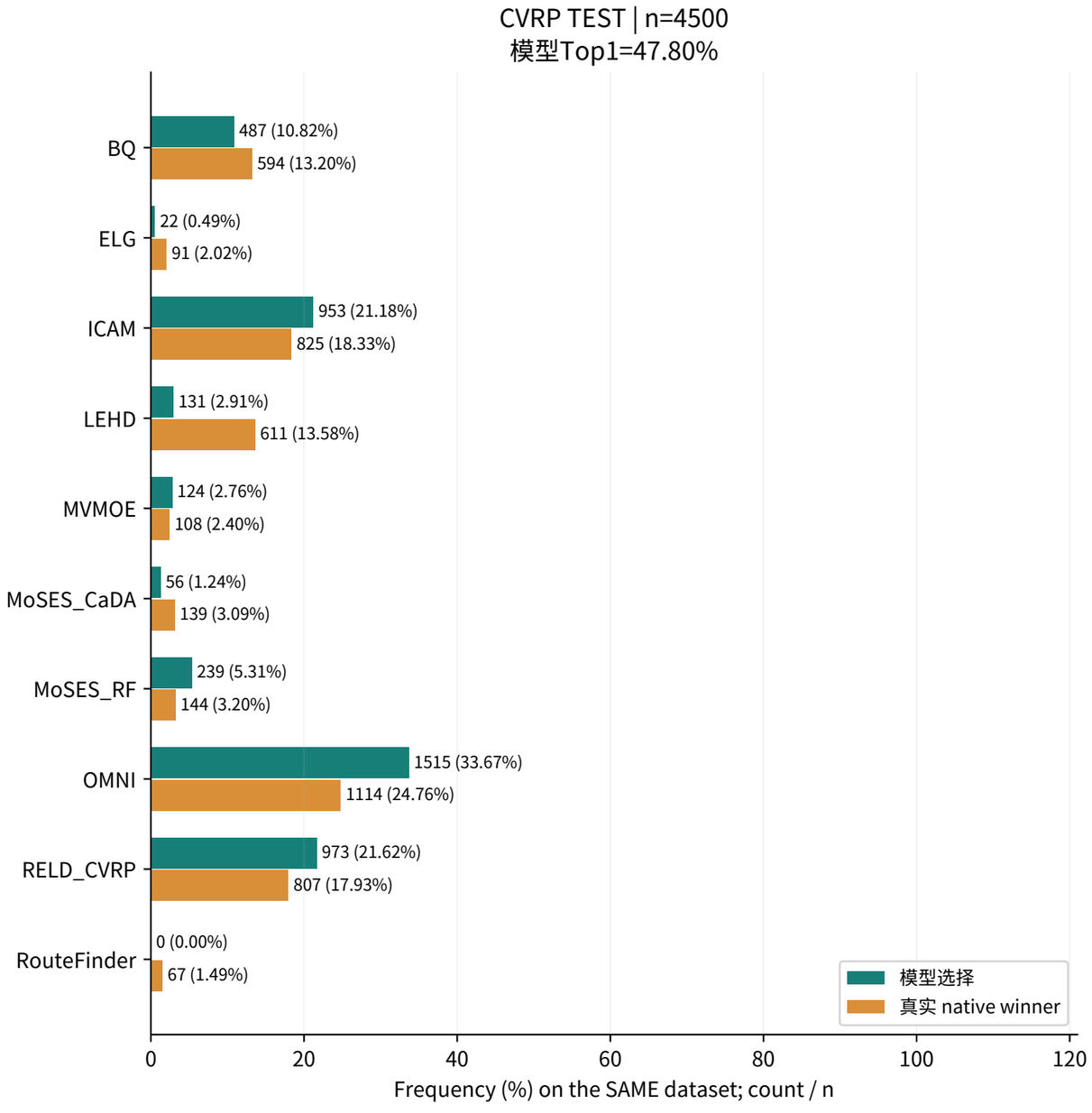
R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

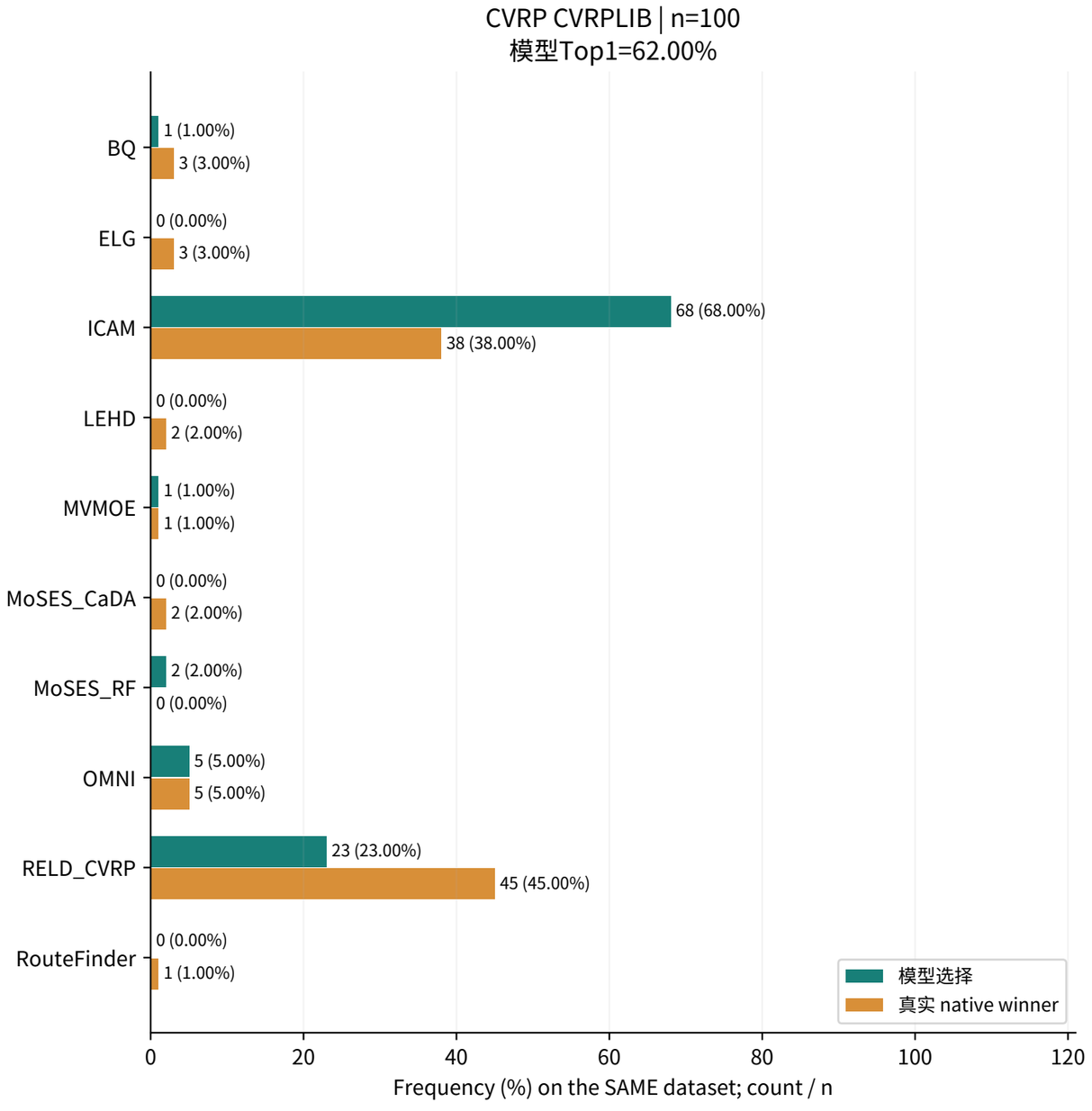




Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

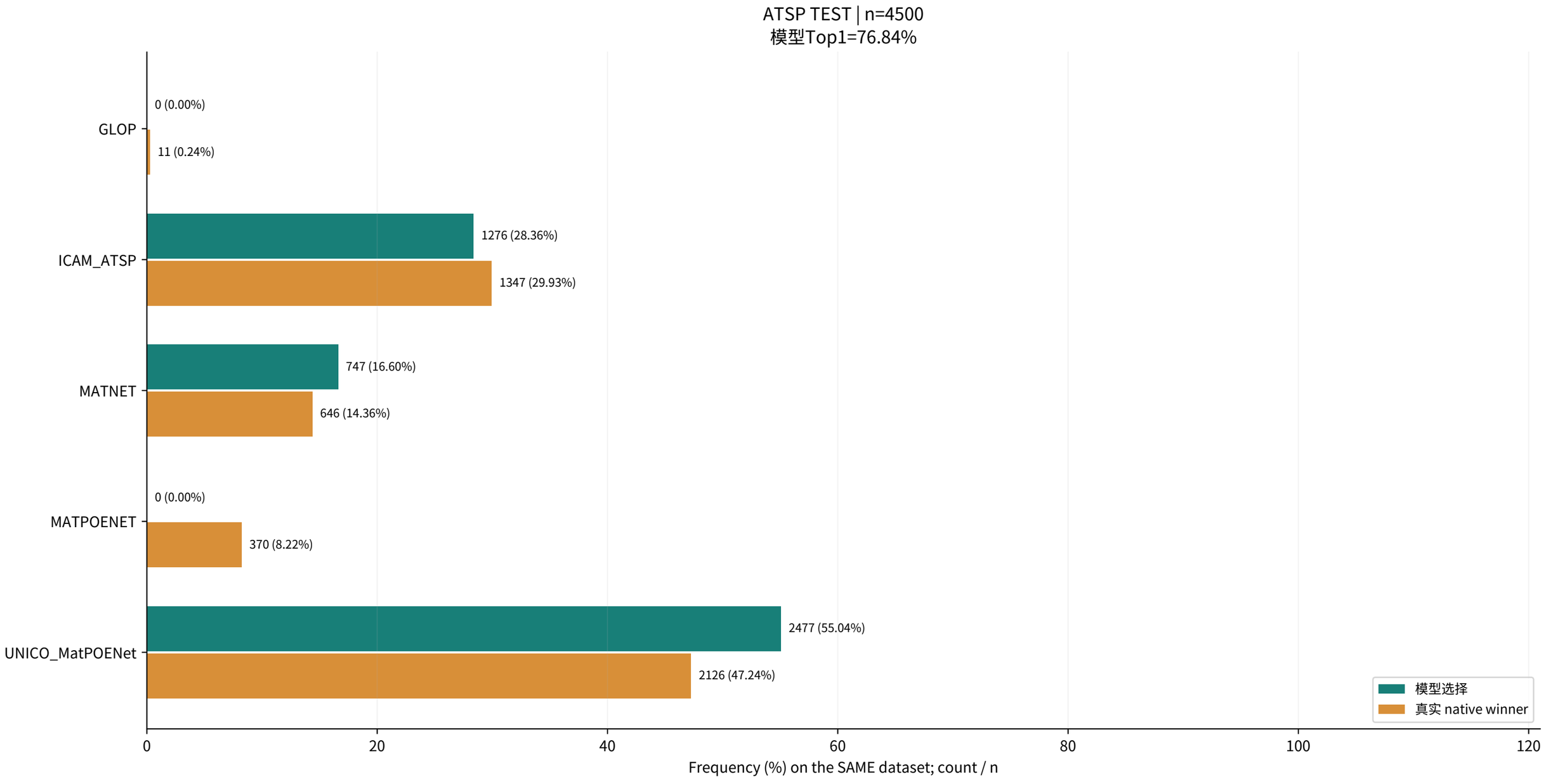
R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018





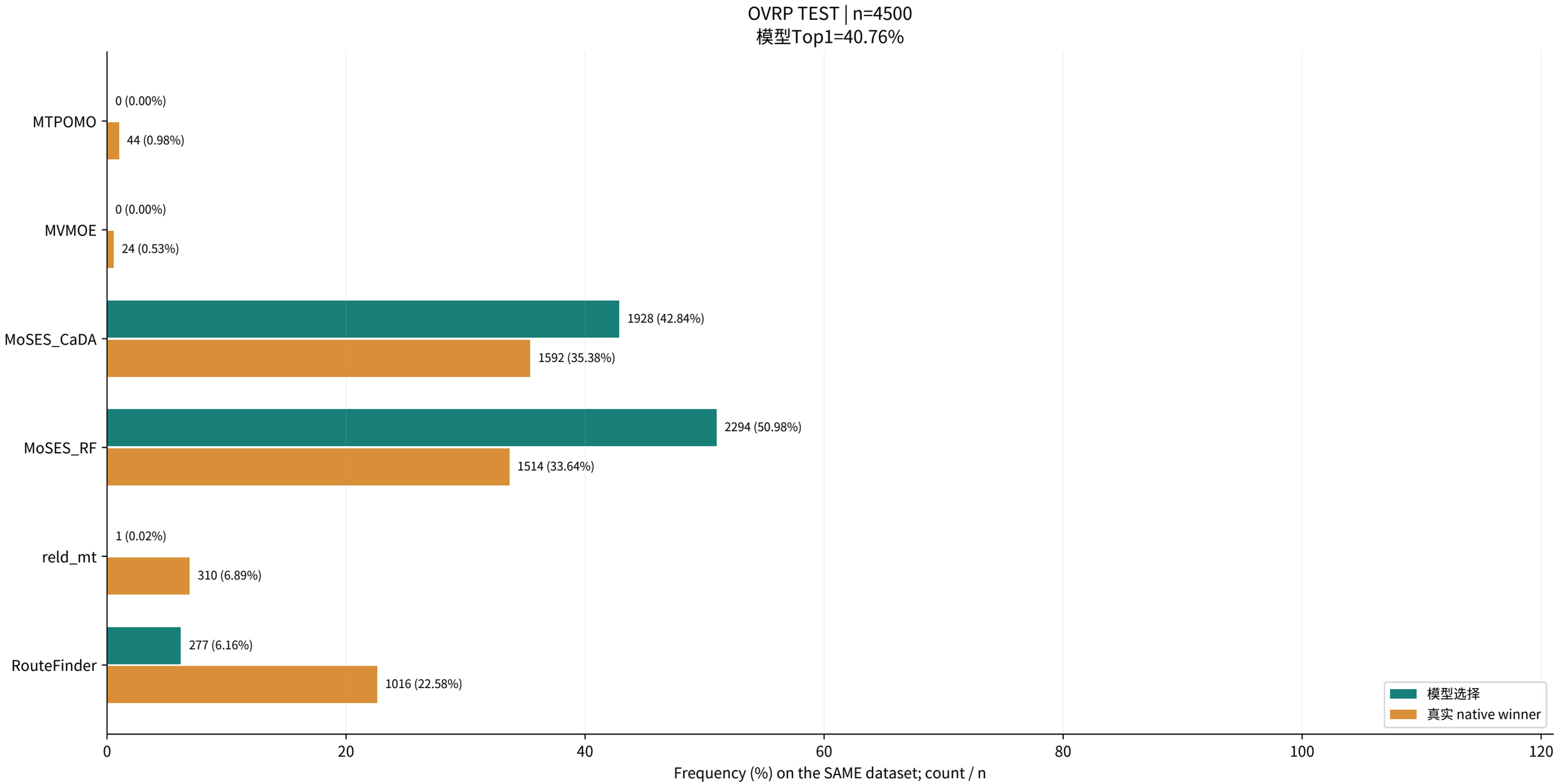
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



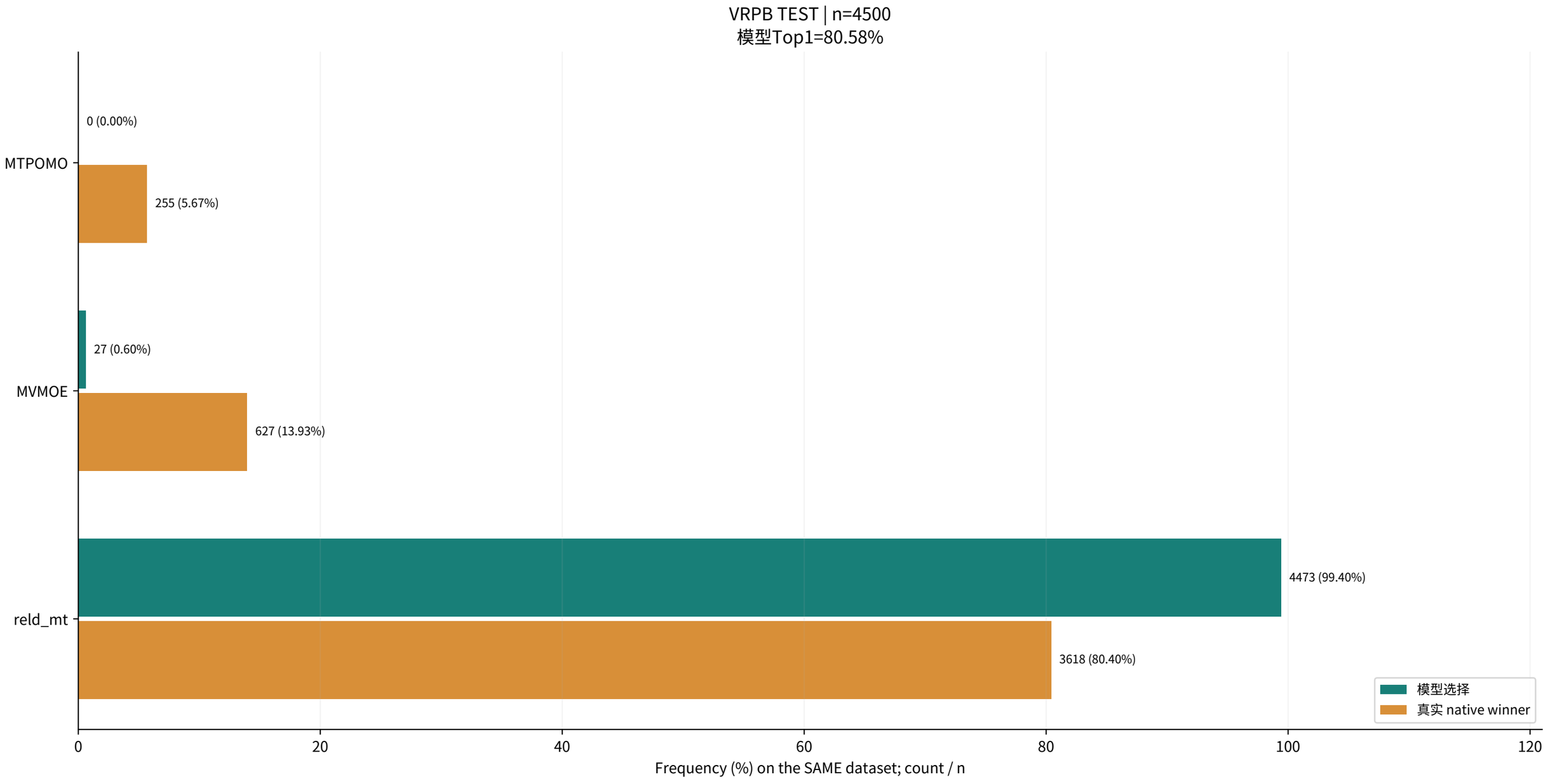
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



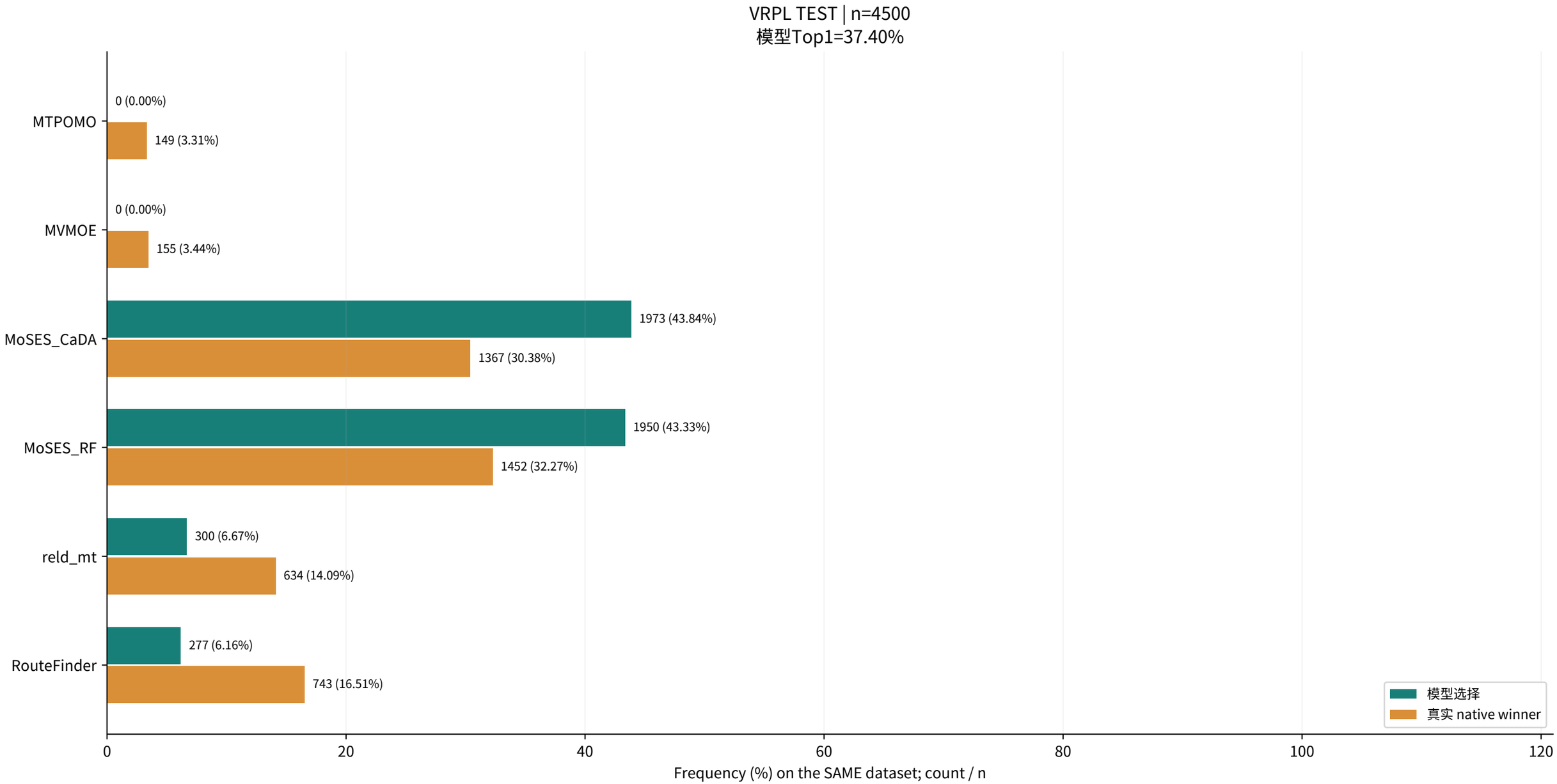
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



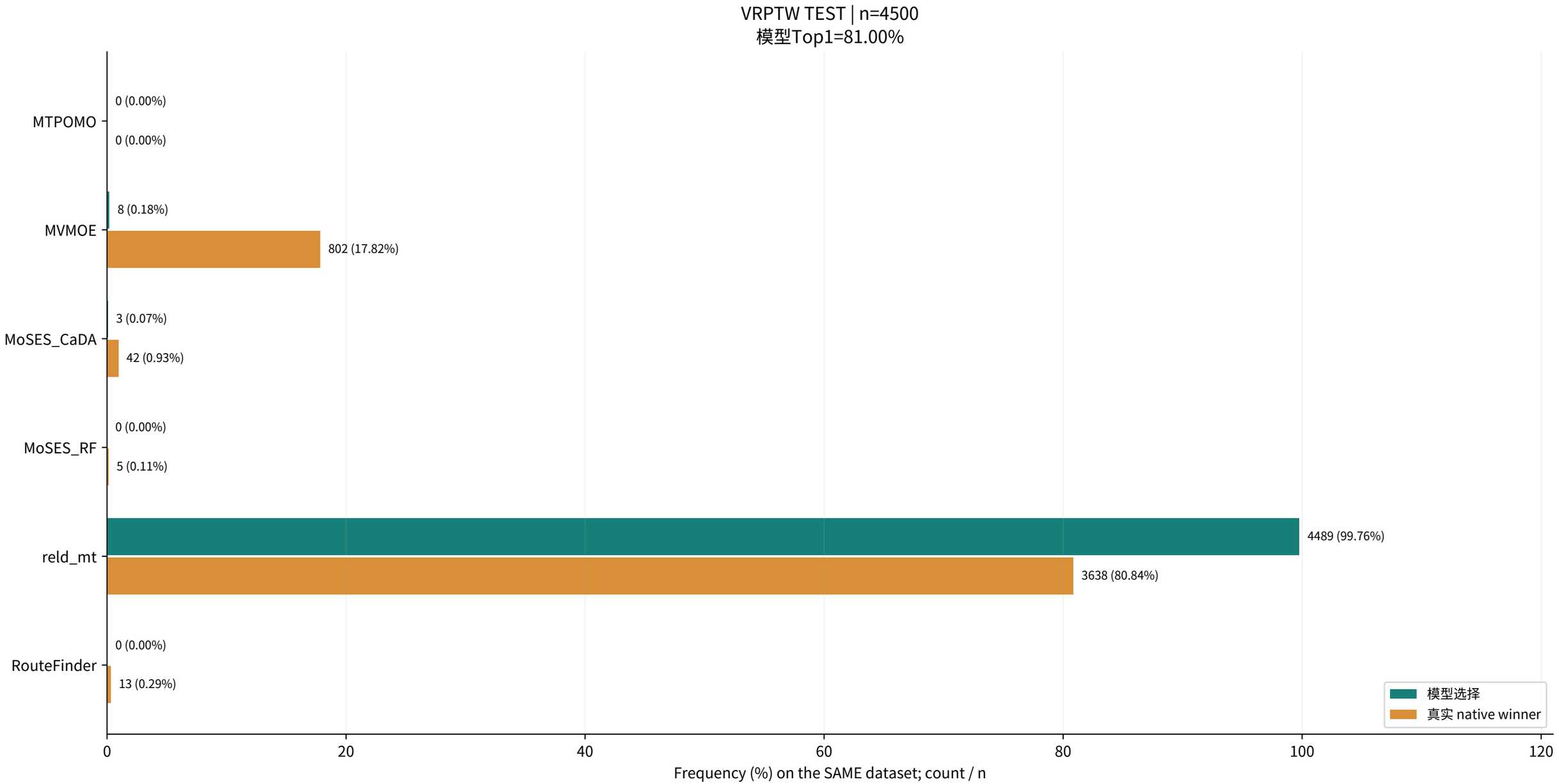
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



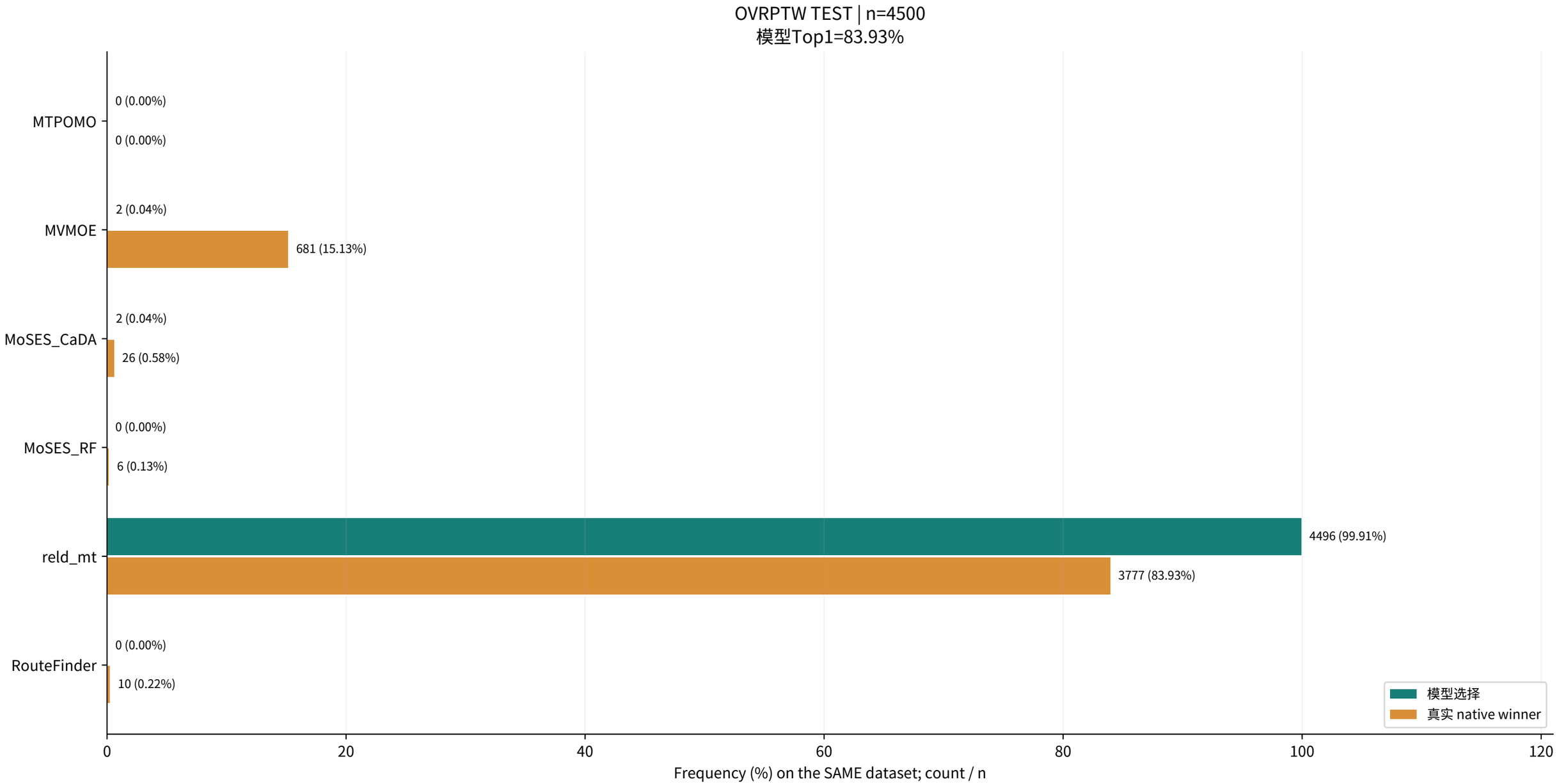
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



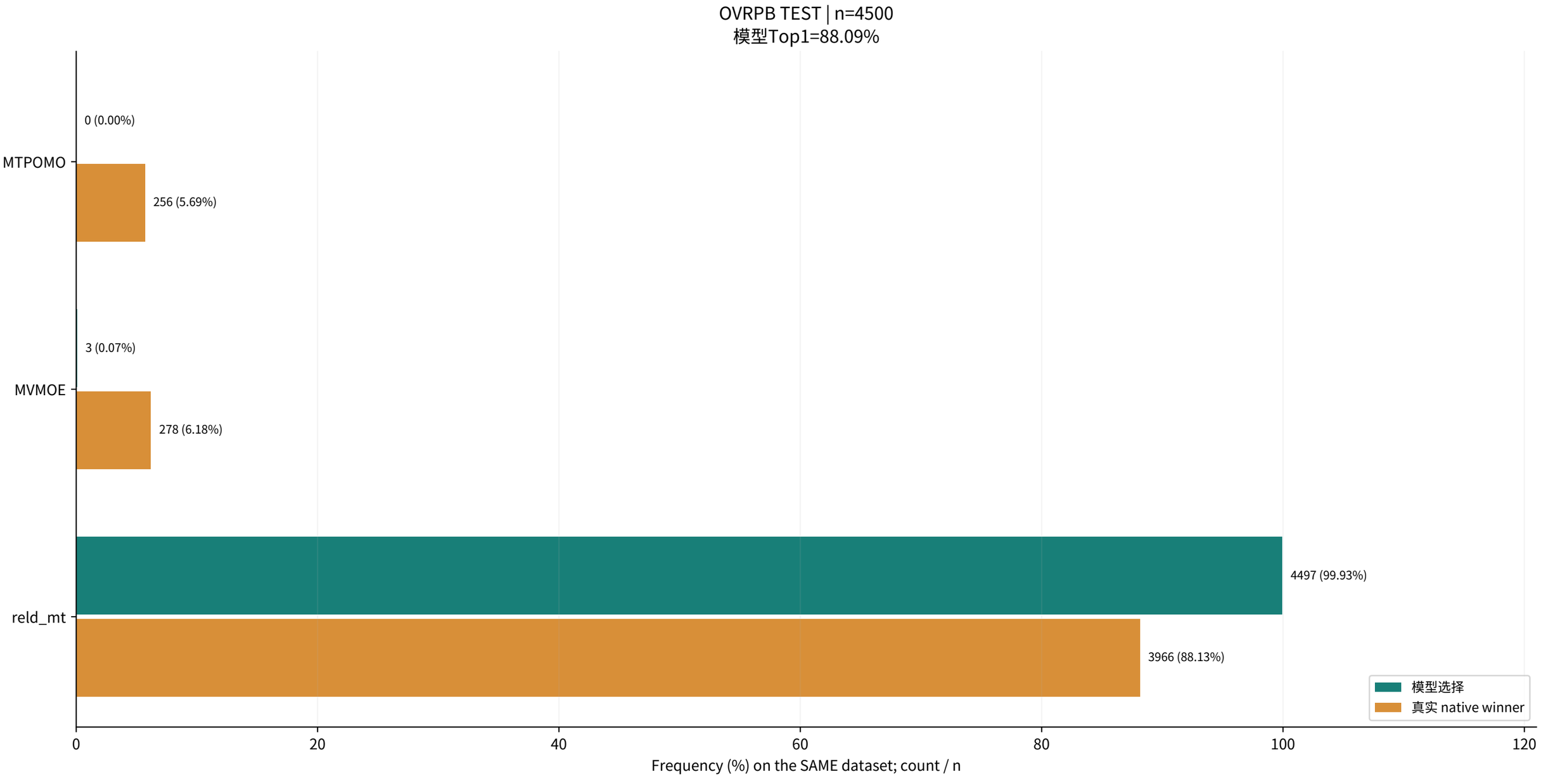
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



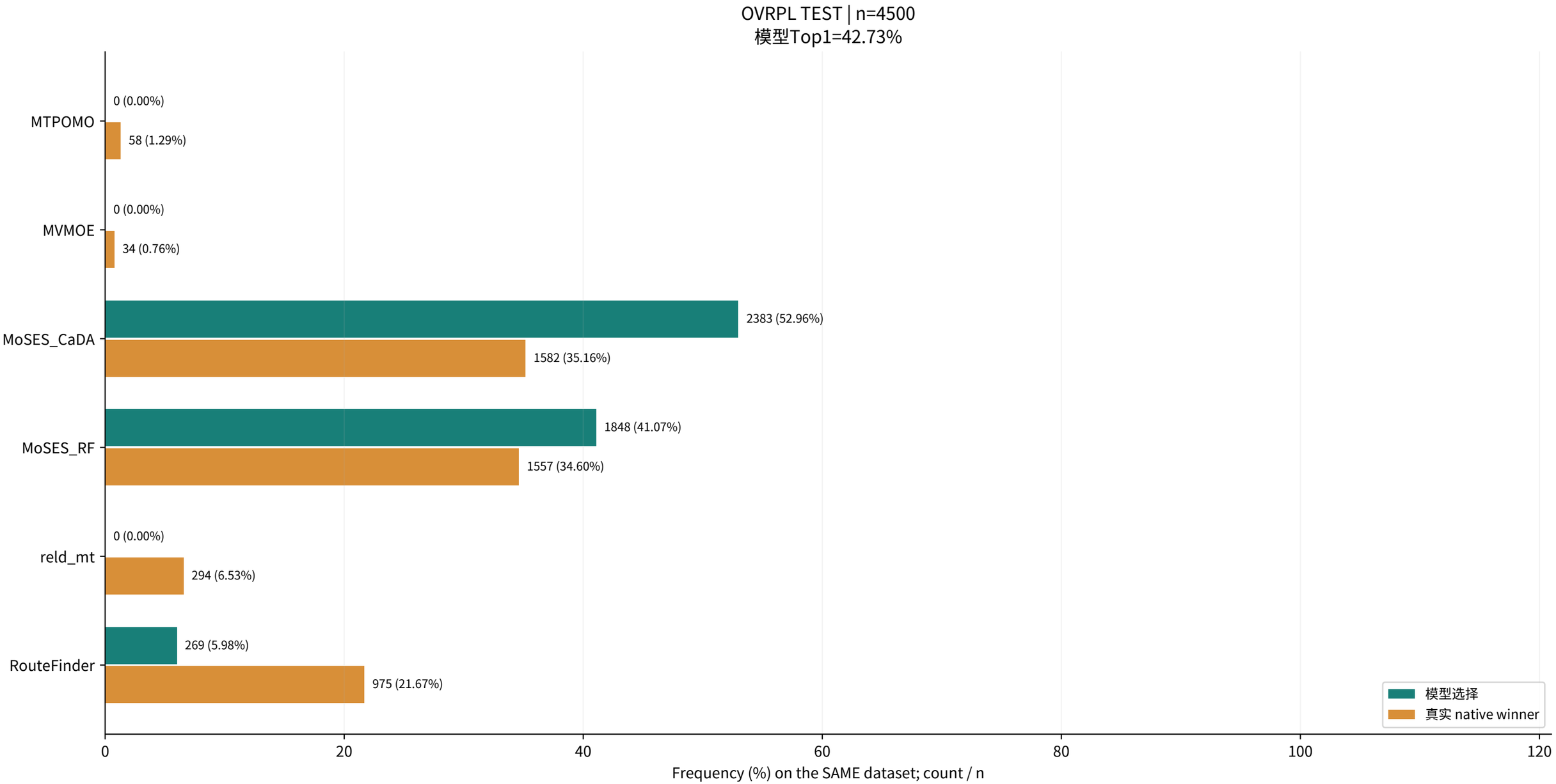
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



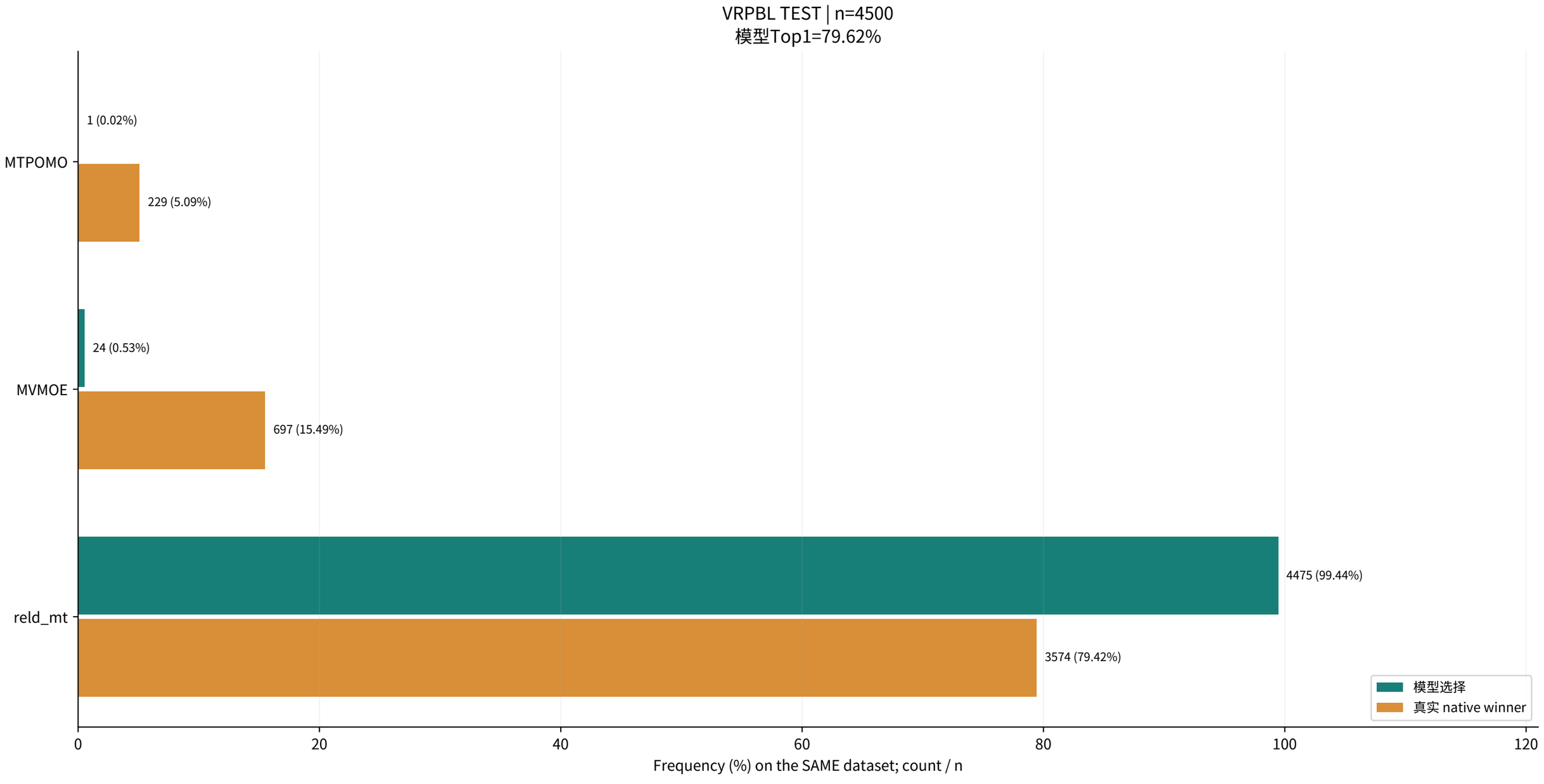
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



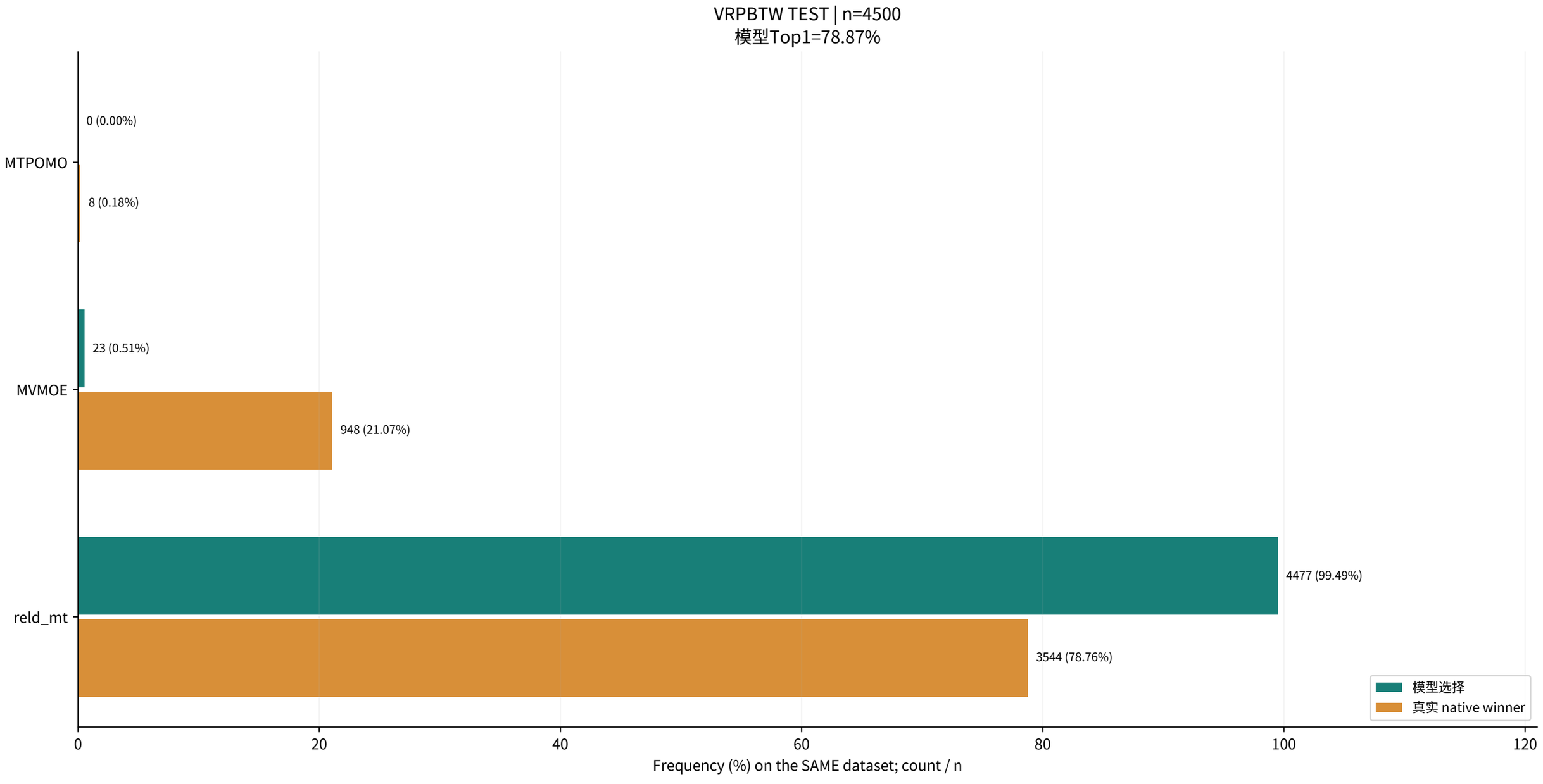
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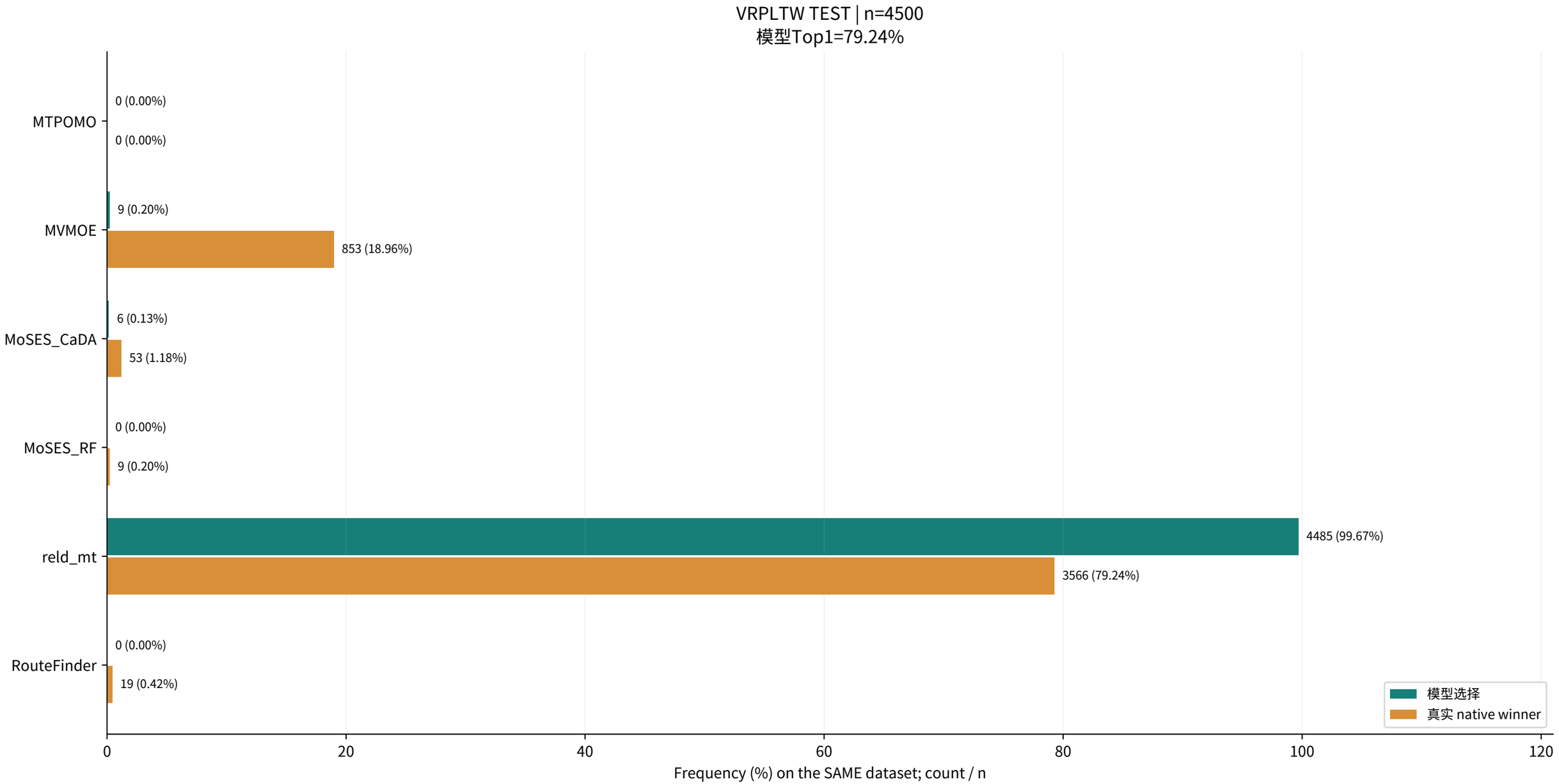
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



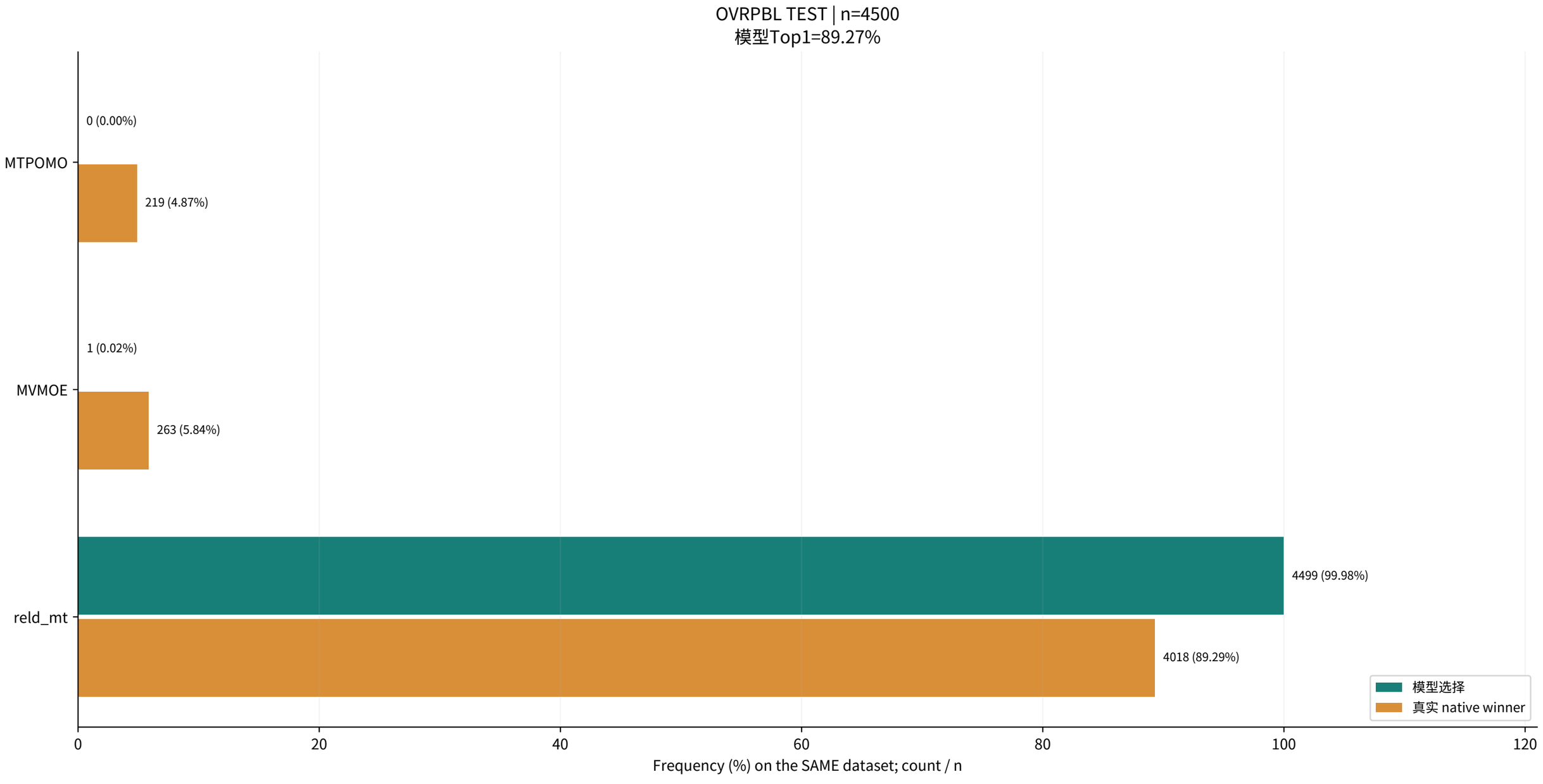
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

R124 | update 5760 | selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018



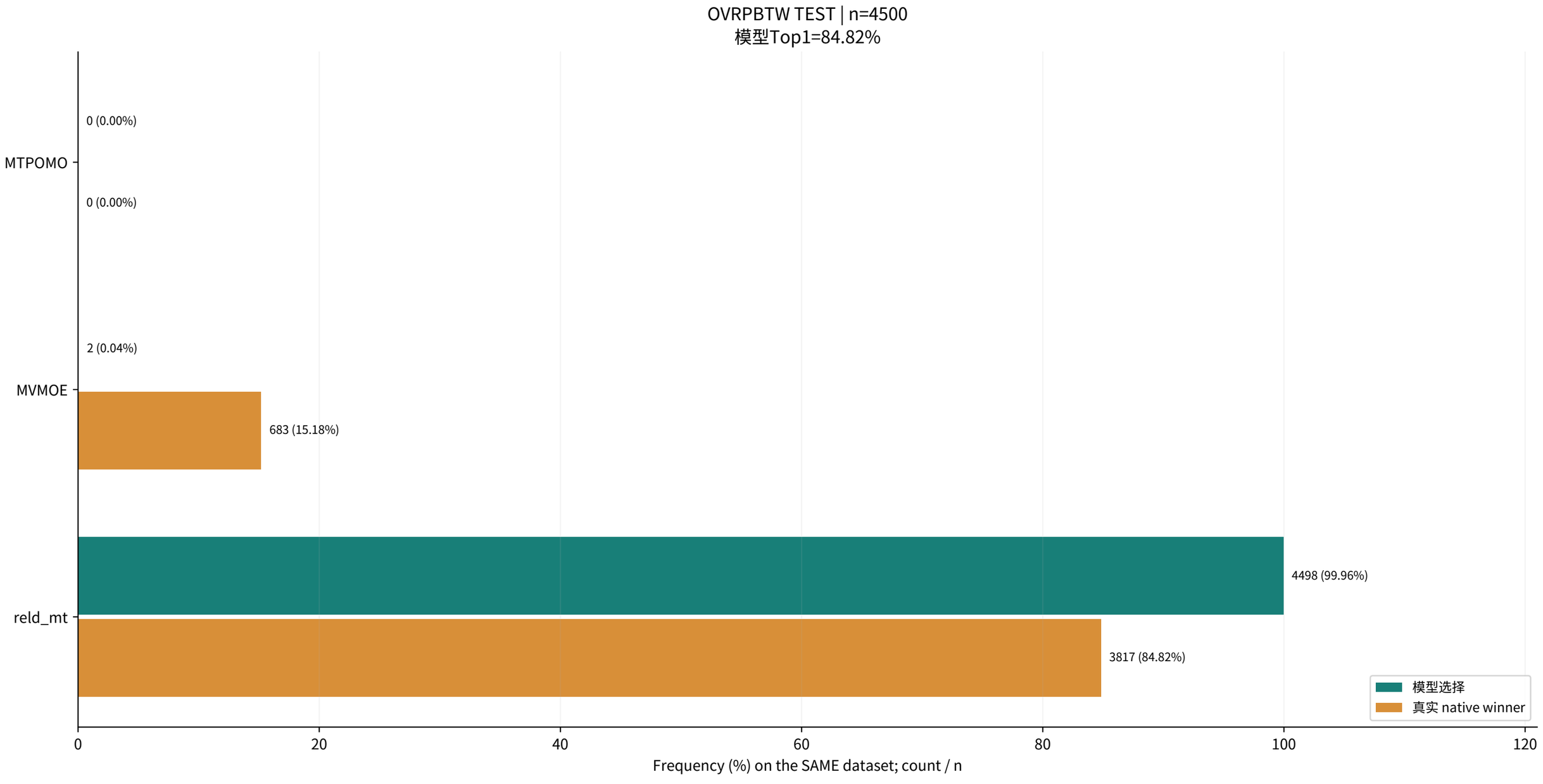
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

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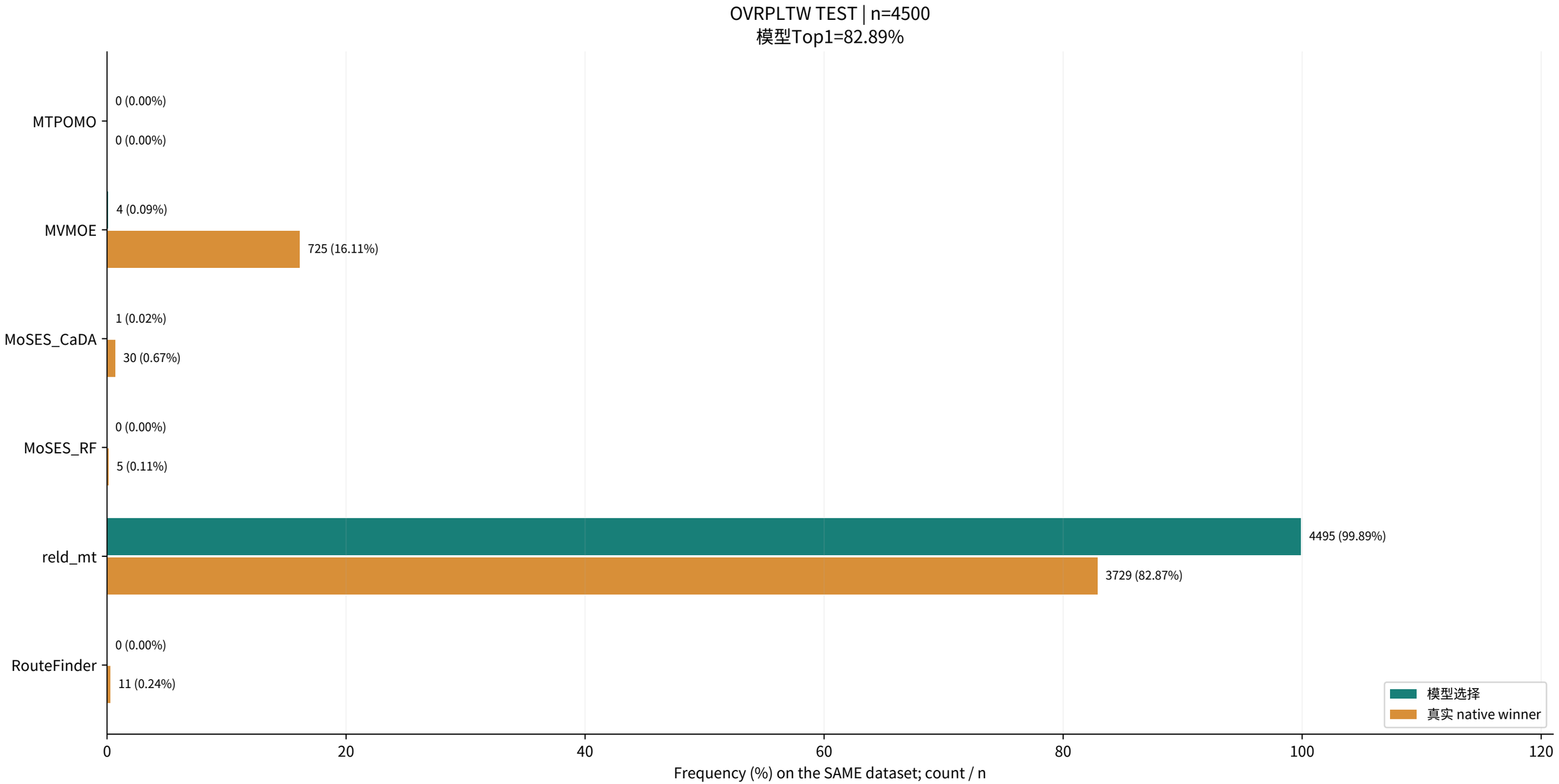
Winner: saved native/FP64 ind, NOT FP32 argmin. Fixed candidate order; true-winner frequency is NOT model accuracy.

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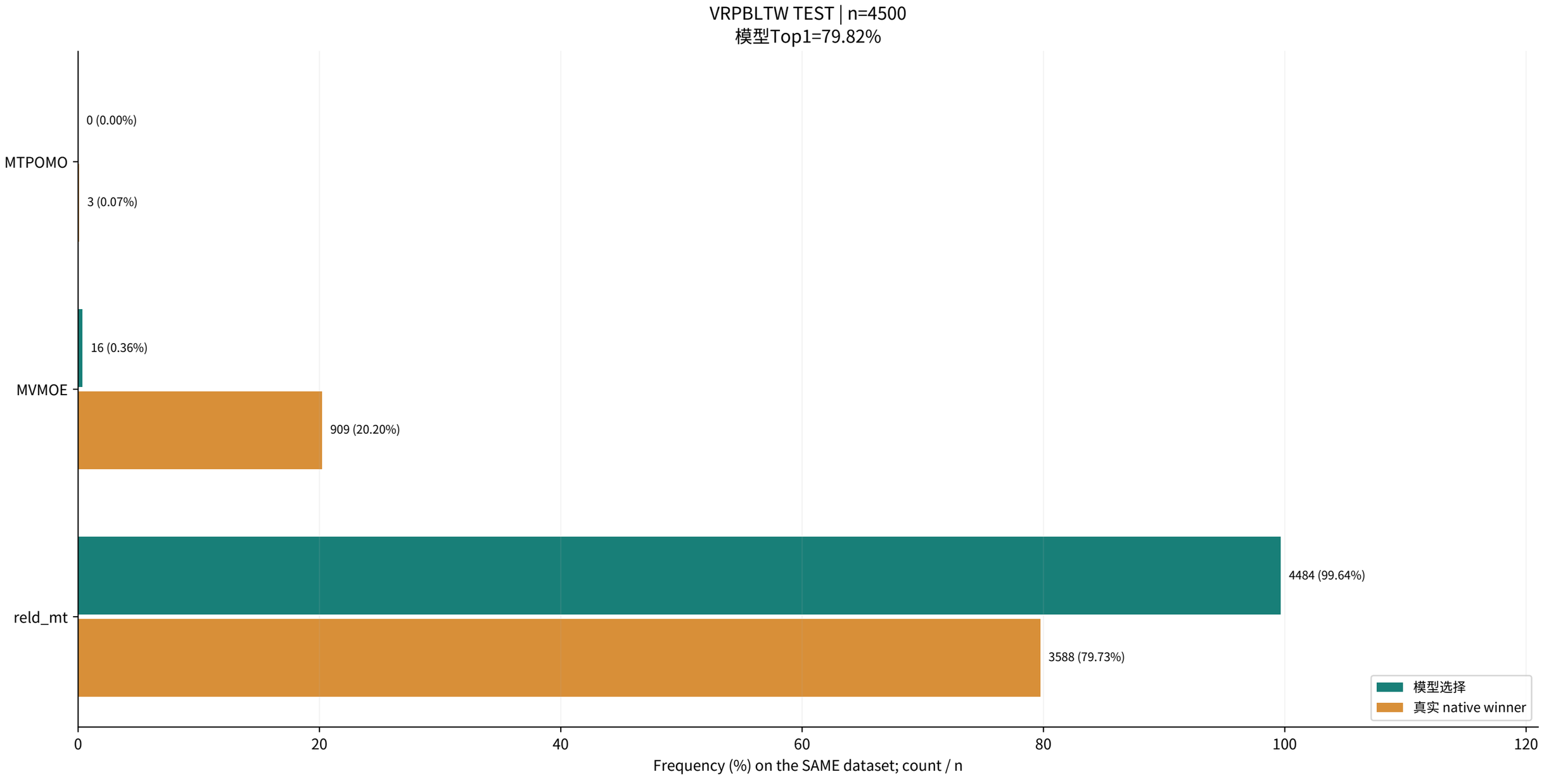
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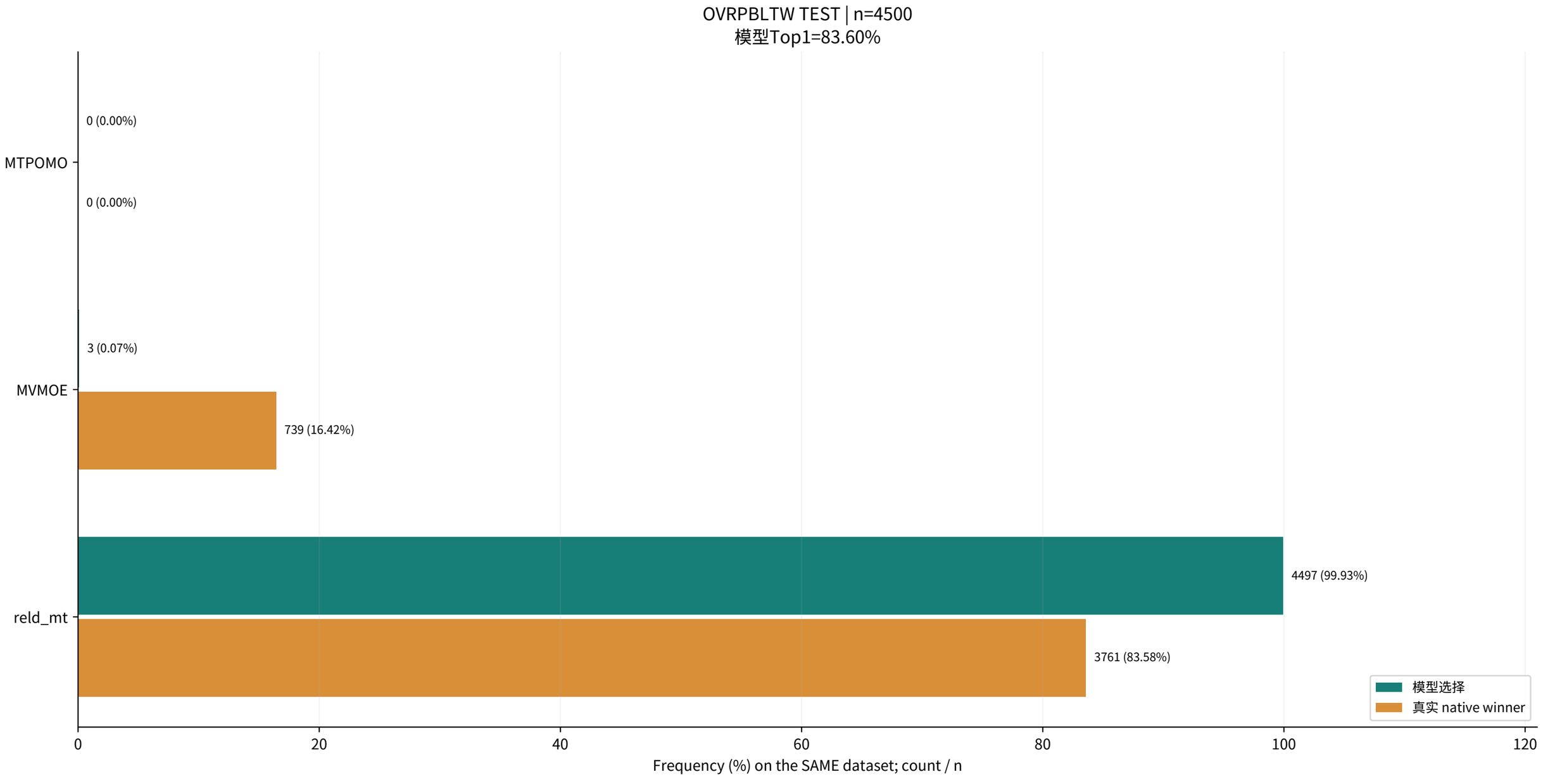
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